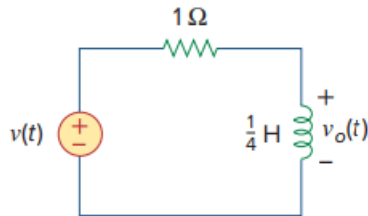


Problem

At what frequency will the output voltage $v_o(t)$ in Fig. 9.39 be equal to the input voltage $v(t)$?

- (a) 0 rad/s
- (b) 1 rad/s
- (c) 4 rad/s
- (d) ∞ rad/s
- (e) none of the above



Step-by-step solution

Step 1 of 1

Refer to Figure 9.39 in the textbook.

The impedance of the inductor is,

$$Z_L = j\omega L$$

As the value of ω is high, the inductor impedance becomes high and acts as open circuit.

At $\omega = 0$ rad/s, $Z_L = 0 \Omega$ and hence the inductor acts as short circuit.

At $\omega = \infty$ rad/s, $Z_L = \infty \Omega$ and hence the inductor acts as open circuit.